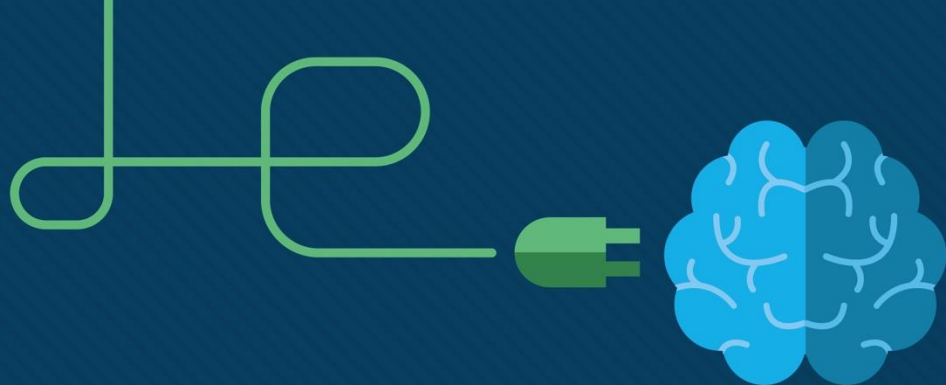




Lab 2.1: Point to Point Single-Area OSPFv2 Configuration

Enterprise Networking, Security, and Automation v7.0
(ENSA)



Outline

- OSPF Router ID
- Point-to-Point OSPF Networks
- Verify Single-Area OSPFv2

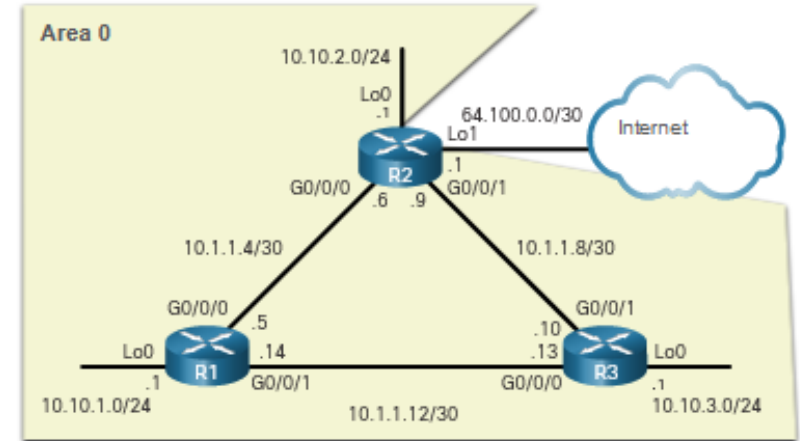
Objectives

- Configure an OSPFv2 router ID.
- Implement single-area OSPFv2 in both point-to-point.
- Verify a single-area OSPFv2 implementation.

OSPF Router ID

OSPF Router ID OSPF Reference Topology

- The routers in the topology have a starting configuration, including interface addresses.
- There is currently no static routing or dynamic routing configured on any of the routers.
- All interfaces on R1, R2, and R3 (except the loopback 1 on R2) are within the OSPF backbone area.
- The ISP router is used as the gateway to the internet of the routing domain.



Router Configuration Mode for OSPF

- OSPFv2 is enabled using the router ospf process-id global configuration mode command.
- The *process-id* value represents a number between 1 and 65,535 and is selected by the network administrator.
- The *process-id* value is *locally significant*.
- It is considered best practice to use the same *process-id* on all OSPF routers.

```
R1(config)# router ospf 10
R1(config-router)# ?
  area                        OSPF area parameters
  auto-cost                  Calculate OSPF interface cost according to
  bandwidth                  bandwidth
  default-information        Control distribution of default information
  distance                   Define an administrative distance
  exit                       Exit from routing protocol configuration mode
  log-adjacency-changes      Log changes in adjacency state
  neighbor                   Specify a neighbor router
  network                    Enable routing on an IP network
  no                         Negate a command or set its defaults
  passive-interface          Suppress routing updates on an interface
  redistribute                Redistribute information from another routing
  protocol
  router-id                  router-id for this OSPF process
R1(config-router)#
```

OSPF Router ID

Router IDs

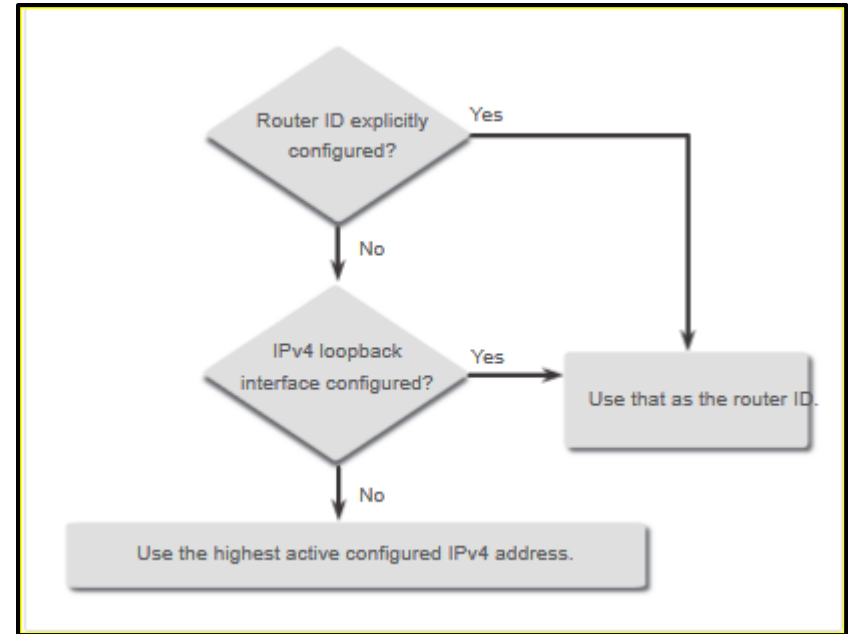
- An OSPF **router ID** is a **32-bit value**, identifier as **IPv4 address**. It is used to uniquely identify an OSPF router, and **all OSPF packets include the router ID** of the originating router.
- **Every router requires a router ID** to participate in an OSPF domain. It can be defined by an administrator or **automatically assigned by the router**.
- The **router ID is used by an OSPF-enabled router** to do the following:
 - Participate in the synchronization of OSPF databases - **During the Exchange State**, the router with the **highest router ID** will send their database descriptor (DBD) packets **first**.
 - Participate in the election of the designated router (DR) - **In a multiaccess LAN environment**, the router with the **highest router ID** is elected the DR. The routing device with the **second highest router ID** is elected the backup designated router (BDR).

OSPF Router ID

Router ID Order of Precedence

Cisco routers derive the router ID based on one of three criteria, in the following preferential order:

1. The router ID is explicitly configured using the *router-id rid* router configuration mode command. This is the recommended method to assign a router ID.
2. The router chooses the *highest IPv4 address of any of configured loopback interfaces*.
3. The router chooses the *highest active IPv4 address of any of its physical interfaces*.



Configure a Loopback Interface as the Router ID

- Instead of relying on physical interface, the router ID can be assigned to a **loopback interface**.
- Typically, the IPv4 address for this type of loopback interface should be configured using a 32-bit subnet mask (255.255.255.255).
- This effectively creates a host route.
- A 32-bit host route would not get advertised as a route to other OSPF routers.
- OSPF does not need to be enabled on an interface for that interface to be chosen as the router ID.

```
R1(config-if)# interface Loopback 1
R1(config-if)# ip address 1.1.1.1 255.255.255.255
R1(config-if)# end
R1# show ip protocols | include Router ID
    Router ID 1.1.1.1
R1#
```

OSPF Router ID

Explicitly Configure a Router ID

In our reference topology the router ID for each router is assigned as follows:

- R1 uses router ID 1.1.1.1
- R2 uses router ID 2.2.2.2
- R3 uses router ID 3.3.3.3

Use the **router-id rid** router configuration mode command to manually assign a router ID.

In the example, the router ID 1.1.1.1 is assigned to R1.

Use the **show ip protocols** command to verify the router ID.

```
R1(config)# router ospf 10
R1(config-router)# router-id 1.1.1.1
R1(config-router)# end
*May 23 19:33:42.689: %SYS-5-CONFIG_I: Configured from console by console
R1# show ip protocols | include Router ID
    Router ID 1.1.1.1
R1#
```

OSPF Router ID

Modify a Router ID

- **After a router selects a router ID**, an active OSPF router does not allow the router ID to be changed until the router is reloaded or the OSPF process is reset.
- **Clearing the OSPF process** is the preferred method to reset the router ID.

```
R1# show ip protocols | include Router ID
Router ID 10.10.1.1
R1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# router ospf 10
R1(config-router)# router-id 1.1.1.1
% OSPF: Reload or use "clear ip ospf process" command, for this to take effect
R1(config-router)# end
R1# clear ip ospf process
Reset ALL OSPF processes? [no]: y
*Jun 6 01:09:46.975: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/1 from FULL to
DOWN, Neighbor Down: Interface down or detached
*Jun 6 01:09:46.981: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/1 from LOADING
to FULL, Loading Done *
R1# show ip protocols | include Router ID
Router ID 1.1.1.1
R1#
```

Point-to-Point OSPF Networks

The network Command Syntax

- Specify the interfaces that belong to a point-to-point network by configuring the **network** command.
- The **basic syntax** for the **network** command is as follows:

```
Router(config-router)# network network-address wildcard-mask area area-id
```

- The **network-address wildcard-mask** syntax is used to enable OSPF on interfaces.
 - Any interfaces on a router that **match this part** of the command are enabled to **send and receive OSPF packets**.
- The **area area-id** syntax **refers to the OSPF area**.
- When **configuring single-area OSPFv2**, the **network** command must be configured with the **same area-id value on all routers**.
 - Although any area ID can be used, it is good practice to use an **area ID of 0** with **single-area OSPFv2**. This convention makes it easier if the network is later altered to support multiarea OSPFv2.

Point-to-Point OSPF Networks

The Wildcard Mask

- The **wildcard mask** is typically the **inverse of the subnet mask** configured on that interface.
- The easiest method for calculating a wildcard mask is to **subtract the network subnet mask from 255.255.255.255**, as shown for /24 and /26 subnet masks in the figure.

Calculating a Wildcard Mask for /24

Subnet Mask	255.255.255.255
	- 255.255.255.000
Wildcard Mask	000.000.000.255

Calculating a Wildcard Mask for /26

Subnet Mask	255.255.255.255
	- 255.255.255.192
Wildcard Mask	000.000.000.063

Subnet mask	Slash notation	Wildcard mask	Rightmost octet(s) of WC in binary
255.255.255.0	/24	0.0.0.255	11111111
255.255.255.128	/25	0.0.0.127	01111111
255.255.255.192	/26	0.0.0.63	00111111
255.255.255.240	/28	0.0.0.15	00001111
255.255.255.248	/29	0.0.0.7	00000111
255.255.254.0	/23	0.0.1.255	00000001.11111111
255.255.252.0	/22	0.0.3.255	00000011.11111111
255.255.248.0	/21	0.0.7.255	00000111.11111111
255.255.0.0	/16	0.0.255.255	11111111.11111111
255.0.0.0	/8	0.255.255.255	11111111.11111111.11111111

Point-to-Point OSPF Networks

The Wildcard Mask

Wildcard 0.0.0.0: Compare all four octets. In other words, the numbers must exactly match.

Wildcard 0.0.0.255: Compare the first three octets only. Ignore the last octet when comparing the numbers.

Wildcard 0.0.255.255: Compare the first two octets only. Ignore the last two octets when comparing the numbers.

Wildcard 0.255.255.255: Compare the first octet only. Ignore the last three octets when comparing the numbers.

Wildcard 255.255.255.255: Compare nothing; this wildcard mask means that all addresses will match the network command.

Command	Logic in Command	Matched Interfaces
network 10.1.0.0 0.0.255.255	Match addresses that begin with 10.1	G0/0.1 G0/0.2 G0/0/0 G0/1/0 G0/2/0
network 10.0.0.0 0.255.255.255	Match addresses that begin with 10	G0/0.1 G0/0.2 G0/0/0 G0/1/0 G0/2/0
network 0.0.0.0 255.255.255.255	Match all addresses	G0/0.1 G0/0.2 G0/0/0 G0/1/0 G0/2/0
network 10.1.13.0 0.0.0.255	Match addresses that begin with 10.1.13	G0/1/0
network 10.1.13.1 0.0.0.0	Match one address: 10.1.13.1	G0/1/0

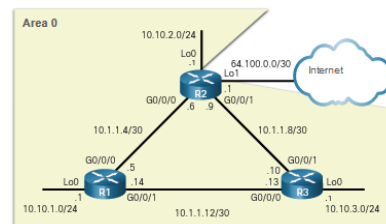
Configure OSPF Using the network Command

Within routing configuration mode, there are *two ways* to *identify the interfaces that will participate in the OSPFv2 routing process*.

- In the first example, the *wildcard mask* identifies the interface based on the *network addresses*.
- Any active interface that is configured with an IPv4 address belonging to that network will participate in the OSPFv2 routing process.
- **Note:** Some IOS versions allow the subnet mask to be entered instead of the wildcard mask. The IOS then converts the subnet mask to the wildcard mask format.

OSPF Single-Area Configuration on R1 Using *network* Command

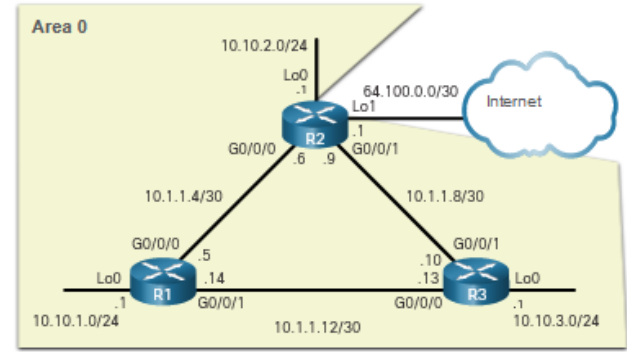
```
R1(config)# router ospf 10
R1(config-router)# network 10.10.1.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.4 0.0.0.3 area 0
R1(config-router)# network 10.1.1.12 0.0.0.3 area 0
R1(config-router)#
```



Configure OSPF Using the network Command (Cont.)

- As an alternative, OSPFv2 can be enabled by specifying the exact interface IPv4 address using a quad zero wildcard mask. Entering **network 10.1.1.5 0.0.0.0 area 0** on R1 tells the router to enable interface Gigabit Ethernet 0/0/0 for the routing process.
- The advantage of specifying the interface is that the wildcard mask calculation is not necessary. Notice that in all cases, the **area** argument specifies area 0.

```
R1(config)# router ospf 10
R1(config-router)# network 10.10.1.1 0.0.0.0 area 0
R1(config-router)# network 10.1.1.5 0.0.0.0 area 0
R1(config-router)# network 10.1.1.14 0.0.0.0 area 0
R1(config-router)#
```



Configure OSPF Using the ip ospf Command

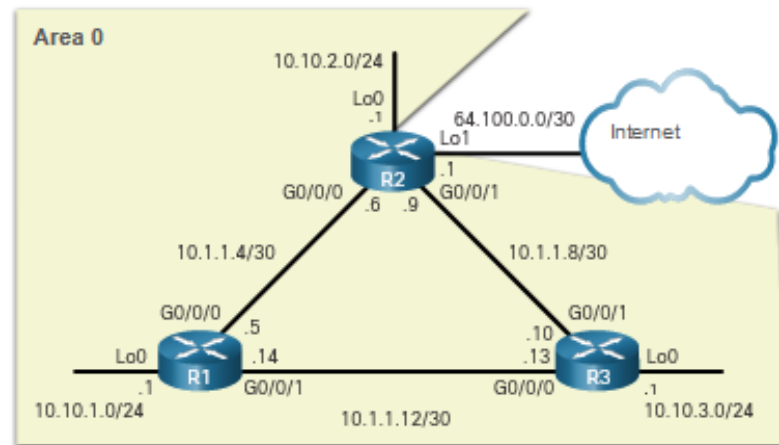
To configure OSPF directly on the interface, use the **ip ospf** interface configuration mode command.

The syntax is as follows:

```
Router(config-if) # ip ospf process-id area area-id
```

- Remove the **network** commands using the **no** form of the command.
- Then go to each interface and configure the **ip ospf** command

```
R1(config)# router ospf 10
R1(config-router)# no network 10.10.1.1 0.0.0.0 area 0
R1(config-router)# no network 10.1.1.5 0.0.0.0 area 0
R1(config-router)# no network 10.1.1.14 0.0.0.0 area 0
R1(config-router)# interface GigabitEthernet 0/0/0
R1(config-if)# ip ospf 10 area 0
R1(config-if)# interface GigabitEthernet 0/0/1
R1(config-if)# ip ospf 10 area 0
R1(config-if)# interface Loopback 0
R1(config-if)# ip ospf 10 area 0
R1(config-if)#
```



Point-to-Point OSPF Networks

Passive Interface

By default, OSPF messages are forwarded out all OSPF-enabled interfaces. However, these messages only need to be sent out interfaces that are connecting to other OSPF-enabled routers.

Sending out unneeded messages on a LAN affects the network in three ways:

- **Inefficient Use of Bandwidth** - Available bandwidth is consumed transporting unnecessary messages.
- **Inefficient Use of Resources** - All devices on the LAN must process and eventually discard the message.
- **Increased Security Risk** - Without additional OSPF security configurations, OSPF messages can be intercepted with packet sniffing software. Routing updates can be modified and sent back to the router, corrupting the routing table with false metrics that misdirect traffic.

Point-to-Point OSPF Networks

Configure Passive Interfaces

- Use the **passive-interface** router configuration mode command *to prevent the transmission of routing messages through a router interface*, but still allow that network to be advertised to other routers.
- The **show ip protocols** command is then used to verify that the interface is listed as passive.

```
R1(config)# router ospf 10
R1(config-router)# passive-interface loopback 0
R1(config-router)# end
R1#
*May 23 20:24:39.309: %SYS-5-CONFIG_I: Configured from console by console
R1# show ip protocols
*** IP Routing is NSF aware ***
(output omitted)
Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Routing on Interfaces Configured Explicitly (Area 0):
    Loopback0
    GigabitEthernet0/0/1
    GigabitEthernet0/0/0
  Passive Interface(s):
    Loopback0
  Routing Information Sources:
    Gateway         Distance      Last Update
    3.3.3.3          110          01:01:48
    2.2.2.2          110          01:01:38
  Distance: (default is 110)
R1#
```

OSPF Point-to-Point Networks

- By default, Cisco routers elect a DR and BDR on Ethernet interfaces, even if there is only one other device on the link.
- You can verify this with the **show ip ospf interface** command.
- The DR/ BDR election process is unnecessary as there can only be two routers on the point-to-point network between R1 and R2.
- Notice in the output that the router has designated the network type as **BROADCAST**.

```
R1# show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 10.1.1.5/30, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
    0                1         no           no           Base
  Enabled by interface config, including secondary ip addresses
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 2.2.2.2, Interface address 10.1.1.6
  Backup Designated router (ID) 1.1.1.1, Interface address 10.1.1.5
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
```

OSPF Point-to-Point Networks (Cont.)

To change this to a point-to-point network, use the interface configuration command **ip ospf network point-to-point** on *all interfaces where you want to disable the DR/BDR election process.*

```
R1(config)# interface GigabitEthernet 0/0/0
R1(config-if)# ip ospf network point-to-point
*Jun 6 00:44:05.208: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0/0 from
FULL to DOWN, Neighbor Down: Interface down or detached
*Jun 6 00:44:05.211: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0/0 from
LOADING to FULL, Loading Done
R1(config-if)# end
R1# show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 10.1.1.5/30, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 1.1.1.1, Network Type POINT_TO_POINT, Cost: 1
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
```

Point-to-Point OSPF Networks

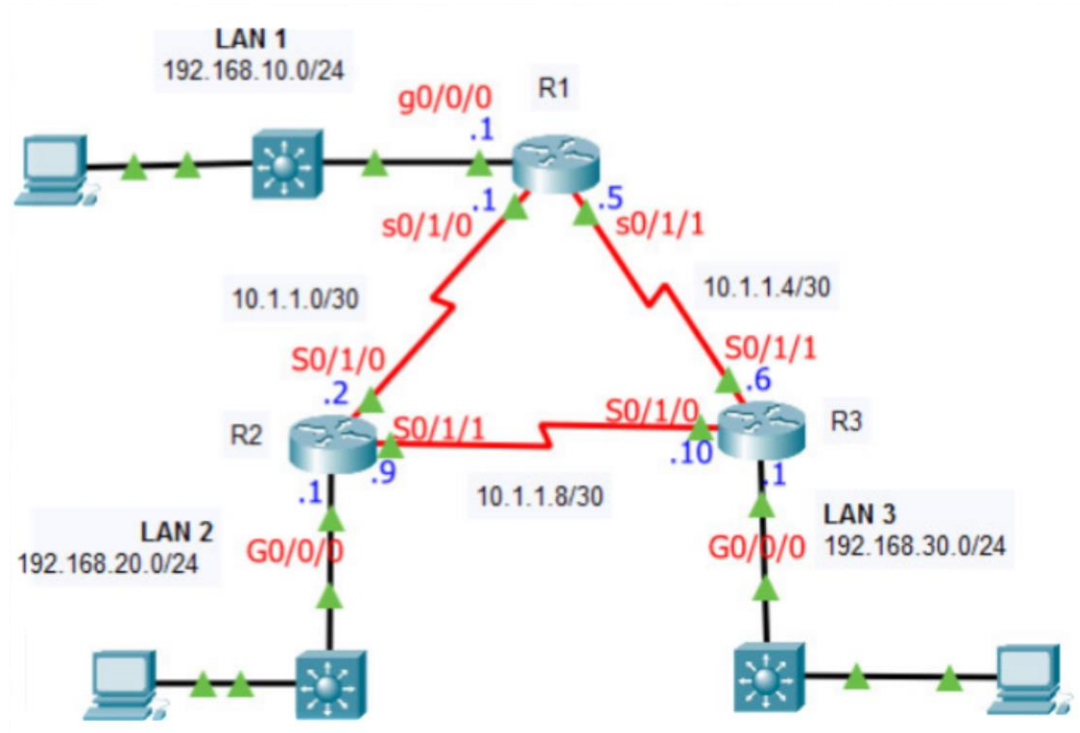
Packet Tracer - Point-to-Point Single-Area OSPFv2 Configuration

In this Packet Tracer activity, you will do the following:

- Explicitly configure router IDs.
- Configure the **network** command on R1 using wildcard mask based on the subnet mask.
- Configure the **network** command on R2 using a quad-zero wildcard mask.
- Configure the **ip ospf** interface command on R3.
- Configure passive interfaces.
- Verify OSPF operation using the **show ip protocols** and **show ip route** commands.

Point-to-Point OSPF Networks

Packet Tracer - Point-to-Point Single-Area OSPFv2 Configuration



Router R1

enable

configure terminal

router ospf 10

router-id 1.1.1.1

#To enter privileged EXEC mode. Short for "enable"

#To enter configuration mode. Short for "configure terminal"

#To start the OSPF routing with process ID 10.

#To specify the router ID

network 192.168.10.0 0.0.0.255 area 0

#This is a single area ospf implementation, therefore we only use area 0

network 10.1.1.1 0.0.0.3 area 0

#To specify the network statement values, IP address and wildcards mask.

network 10.1.1.4 0.0.0.3 area 0

passive-interface g0/0/0

#To prevent the OSPF process from sending unnecessary routing protocol traffic

Router R2

enable

Configure terminal

router ospf 10

router-id 2.2.2.2

network 192.168.20.1 0.0.0.0 area 0

network 10.1.1.2 0.0.0.0 area 0

network 10.1.1.9 0.0.0.0 area 0

passive-interface g0/0/0

Router R3

```
router ospf 10
router-id 3.3.3.3
passive-interface g0/0/0
int g0/0/0
ip ospf 10 area 0
int s0/1/0
ip ospf 10 area 0
int s0/1/1
ip ospf 10 area 0
```

Verify passive interface

R1

```
R1#show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.10.0 0.0.0.255 area 0
    10.1.1.0 0.0.0.3 area 0
    10.1.1.4 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:17:07
    2.2.2.2          110          00:17:40
    3.3.3.3          110          00:17:07
  Distance: (default is 110)
```

R1

```
R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/1/0
L       10.1.1.1/32 is directly connected, Serial0/1/0
C       10.1.1.4/30 is directly connected, Serial0/1/1
L       10.1.1.5/32 is directly connected, Serial0/1/1
O       10.1.1.8/30 [110/128] via 10.1.1.2, 00:16:06, Serial0/1/0
          [110/128] via 10.1.1.6, 00:16:06, Serial0/1/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
O       192.168.20.0/24 [110/65] via 10.1.1.2, 00:18:30, Serial0/1/0
O       192.168.30.0/24 [110/65] via 10.1.1.6, 00:16:06, Serial0/1/1
```

R2

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/1/0
L       10.1.1.2/32 is directly connected, Serial0/1/0
O       10.1.1.4/30 [110/128] via 10.1.1.1, 00:22:36, Serial0/1/0
          [110/128] via 10.1.1.10, 00:22:36, Serial0/1/1
C       10.1.1.8/30 is directly connected, Serial0/1/1
L       10.1.1.9/32 is directly connected, Serial0/1/1
O       192.168.10.0/24 [110/65] via 10.1.1.1, 00:24:59, Serial0/1/0
        192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0
O       192.168.30.0/24 [110/65] via 10.1.1.10, 00:23:09, Serial0/1/1
```

R3

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O       10.1.1.0/30 [110/128] via 10.1.1.9, 00:23:27, Serial0/1/0
        [110/128] via 10.1.1.5, 00:23:27, Serial0/1/1
C       10.1.1.4/30 is directly connected, Serial0/1/1
L       10.1.1.6/32 is directly connected, Serial0/1/1
C       10.1.1.8/30 is directly connected, Serial0/1/0
L       10.1.1.10/32 is directly connected, Serial0/1/0
O       192.168.10.0/24 [110/65] via 10.1.1.5, 00:23:27, Serial0/1/1
O       192.168.20.0/24 [110/65] via 10.1.1.9, 00:24:00, Serial0/1/0
        192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0/0
```